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Report Title: SEG3125 Assignment 4 — E-Commerce Page Report

Goal

a. Name and type of e-commerce

The name of my e-commerce site is MN Powerlifting Store. It is a powerlifting equipment and clothing e-commerce website. The site sells training and competition products such as powerlifting belts, knee sleeves, wrist wraps, competition singlets, and competition t-shirts. The main target users are strength athletes and beginner-to-intermediate powerlifters who want to browse, compare, and purchase lifting products for training or meet day.

b. Inspiration sites

1. <https://sbdapparel.ca/>

The first inspiration site is SBD Apparel Canada. I used this site as inspiration for the visual style of my e-commerce site. SBD uses a strong black, white, and red color theme, which creates a serious, professional, and competitive powerlifting mood. This influenced my own design choice because I wanted MN Powerlifting Store to feel bold and athletic instead of casual. I also used a similar high-contrast style to make headings, buttons, product cards, and navigation elements stand out clearly.

2. <https://a7canada.com/>

The second inspiration site is A7 Canada. I used this site as inspiration because it presents powerlifting gear and apparel with a strong athletic brand image. A7 separates different product types clearly, such as equipment and clothing, which influenced my navigation and product organization. In my site, users can browse all products or narrow their search by Equipment and Clothing, then further explore product types such as belts, knee sleeves, and competition clothing.

3. <https://innerstrengthproducts.ca/>

Inner Strength Products was used as inspiration because it is a Canadian strength training store that sells powerlifting-related products. This inspired my decision to build a powerlifting-focused store instead of a general fitness shop. It also influenced my product categories, especially the separation between equipment and competition clothing.

Reflection / Design

(A) Interactive Process / System Image Design

a. Follow instruction process

For the following instruction process, I designed a checkout flow that guides the user step by step. After a user adds a product to the cart, they can review the cart, enter personal information, enter payment information, review the order, and finally receive an order confirmation. To provide feedback, I used a checkout stepper with four stages: Cart, Personal Info, Payment, and Confirmation. This

shows the user where they currently are, what has already been completed, and what still needs to be done. I also included buttons such as “Back to Cart,” “Continue to Payment,” “Review Order,” and “Place Order” so the user can clearly understand the next action.

b. Explore process / faceted search

For the exploration process, I used a faceted search structure based on product categories and product types. Users can begin from “All Products,” then narrow their browsing by choosing “Equipment” or “Clothing.” Under Equipment, users can filter by Belts, Knee Sleeves, and Wrist Wraps. Under Clothing, users can filter by Competition Singlets and Competition T-Shirts. This structure matches the way powerlifters usually think about shopping for gear: first by general category, then by the specific product type they need. The product cards also show information such as category, type, variation, focus, and price, so users can compare products without needing to remember details from other pages.

c. Communicate process

For the communication process, I placed a short survey after the order confirmation page. I decided to put the survey after checkout because it does not interrupt the user while they are browsing products or completing payment. The survey is simple and asks about the overall experience, whether it was easy to find products, and any comments the user wants to leave. This makes the survey less intrusive because it appears only after the main shopping task is complete. It is also engaging because the language is positive and explains that the feedback helps improve the store experience.

(B) Verbal Communication Design

a. Writer / Reader Model

The Writer/Reader model of my site is a professional powerlifting store speaking to strength athletes and beginner-to-intermediate powerlifters. The writer voice is confident, direct, and supportive. The reader is a user who wants to find reliable powerlifting equipment or clothing for training and competition. I kept this model consistent by using clear product language, direct calls to action, and short informative descriptions. For example, the site uses phrases such as “Equipment and clothing for strength athletes,” “Shop Now,” and “Everything you need to train hard, lift strong, and compete with confidence.” These phrases make the site feel like a serious athletic brand while still being easy to understand.

b. Communicative Purposes

Incite to action:

I used action-oriented language to encourage users to shop and complete the buying process. Examples include “Shop Now,” “View All Products,” “Add to Cart,” and “Ready to complete your setup?” These are short imperative or promotional phrases that guide users toward purchasing products.

Inform:

I used informative language in the product cards and product detail pages. Each product includes its name, category, type, variation, focus, price, and a short description. For example, the 10mm Powerlifting Belt is described as “a firm 10mm belt for lifters who want support without losing setup comfort.” This helps users understand the product before adding it to the cart.

Engage in a connection:

I used friendly and positive language in the survey section to create a connection with the user. The survey says, “Help us improve the store” and asks questions such as “Was it easy to find products?” This makes the user feel that their feedback matters. The message after submission also thanks the user for their feedback, which supports a more personal and engaging communication style.

High-Fidelity Prototype

a. Visual Design Choices

For the high-fidelity prototype, I wanted the e-commerce site to look consistent with my previous MN Powerlifting service site and memory game. I used a black, white, and red color palette to create a strong and serious powerlifting mood. The black background gives the site a professional and intense feeling, the red is used for important brand elements and action buttons, and the white space helps keep the content readable. For typography, I used Michroma for the brand name and main headings because it matches the strong and modern style of the MN Powerlifting brand. I used Squada One for product names and card headings because it creates a bold athletic feeling. For body text, product descriptions, forms, and smaller information, I used Source Sans Pro because it is easier to read.

The layout is designed to support shopping and product exploration. The home page starts with a large hero image and a clear “Shop Now” button to guide users into the store. The product pages use card-based layouts so that users can scan product images, names, categories, variations, prices, and action buttons quickly. I also used consistent red buttons for major actions such as “Shop Now,” “Add to Cart,” “Checkout,” and “Place Order,” so users can easily recognize the next step.

The navigation design also supports the e-commerce goal. The main navigation includes About Me, E-commerce Site, Cart, and Purchase History. Inside the store, the sub navigation allows users to browse All Products, Equipment, Clothing, and product subcategories such as Belts, Knee Sleeves, Wrist Wraps, Competition Singlets, and Competition T-Shirts. This makes the faceted search visible and easy to use without adding a search box.

The checkout pages use a clear step-by-step structure with Cart, Personal Info, Payment, and Confirmation. This helps users understand where they are in the purchase process and what remains to be completed. The survey is placed after the order confirmation so it does not interrupt the shopping or checkout experience.

b. Portfolio Link

Link to UI

ziye-seg3125-project.netlify.app

Link to GitHub Repository

https://github.com/Ziye-Zhou/SEG3125_Project

Heuristic Evaluation

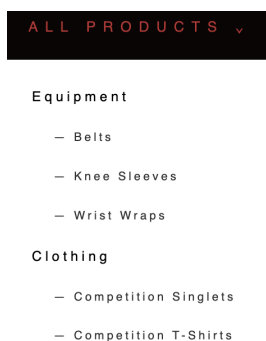
1. Visibility of System Status

My site shows the user's current status during the checkout process. The checkout stepper includes Cart, Personal Info, Payment, and Confirmation, so users can see where they are in the process and what still needs to be completed. The cart page also shows the quantity and total price, so users can understand the current state of their order.

The screenshot shows a checkout interface with a dark background. At the top, the word "Checkout" is written in a light blue font. Below it is a progress bar with four steps: "1. Cart", "2. Personal Info", "3. Payment", and "4. Confirmation". The "2. Personal Info" step is currently active. Below the progress bar are several input fields: "Full Name" with the value "ziye", "Email" with the value "ziye@gmail.com", "Phone" with the value "123", and "Shipping Address" with the value "123". At the bottom, there are two buttons: "Back to Cart" and "Continue to Payment".

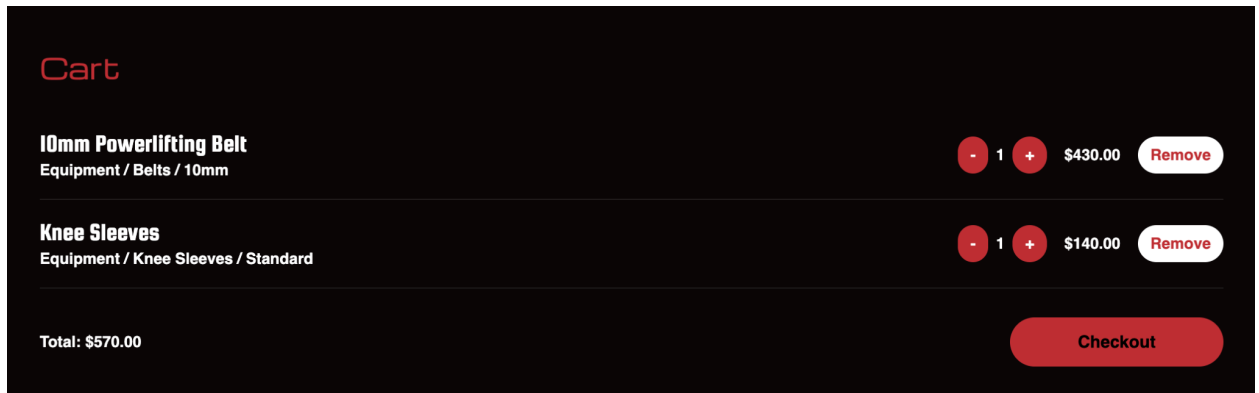
2. Match Between the System and the Real World

The product categories use terms that powerlifting users would recognize, such as Equipment, Clothing, Belts, Knee Sleeves, Wrist Wraps, Competition Singlets, and Competition T-Shirts. These words match the language used by strength athletes when shopping for lifting gear. The product descriptions also use familiar terms such as squat, deadlift, bench, meet day, and competition support.



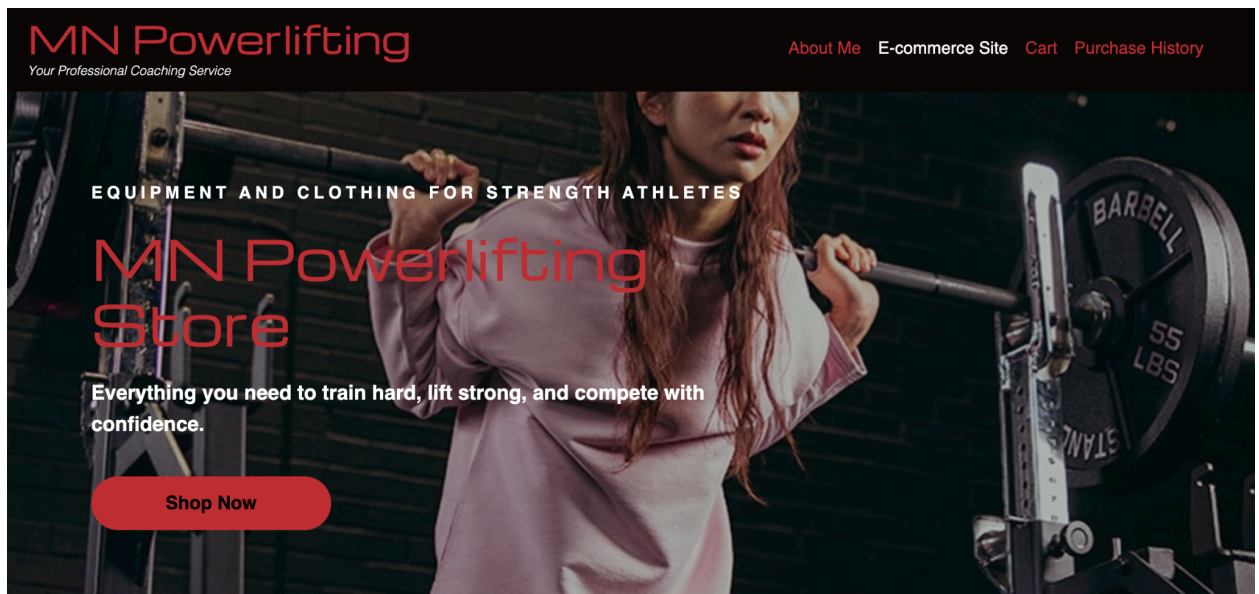
3. User Control and Freedom

Users can control their shopping path in several ways. They can return to the Home Page, browse All Products, filter by Equipment or Clothing, remove items from the cart, go back to the cart during checkout, or continue shopping after order confirmation. These options allow users to recover from choices or change their mind without feeling trapped.



4. Consistency and Standards

The site uses consistent colors, fonts, and button styles across all pages. Red buttons are used for important actions such as Shop Now, Add to Cart, Checkout, and Place Order. The black, white, and red color system is also consistent with the MN Powerlifting brand used in my previous service site and memory game. This helps users understand which elements are clickable and important.



5. Error Prevention

The checkout form uses required fields for personal information and payment information. This prevents users from continuing without entering necessary details such as name, email, phone number, shipping address, card name, card number, expiry date, and CVV. The checkout flow also asks users to review the order before

placing it, which helps prevent mistakes before final confirmation.

Email

! Please include an '@' in the email address. 'ziye' is missing an '@'.

Phone

Shipping ! Please match the requested format.
Phone number must be exactly 10 digits.

Card Number

CVV ! Please match the requested format.
Card number must be exactly 16 digits.

Expiry

! Please match the requested format.
Expiry must be in MM/YY format, for example 08/28.

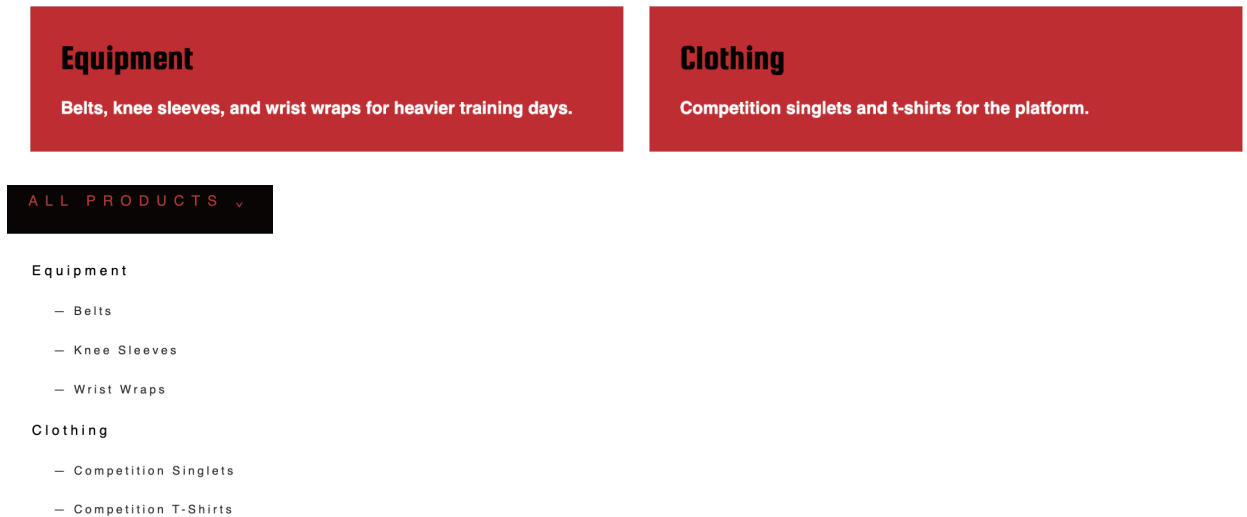
CVV

! Please match the requested format.
CVV must be exactly 3 digits.

6. Recognition Rather Than Recall

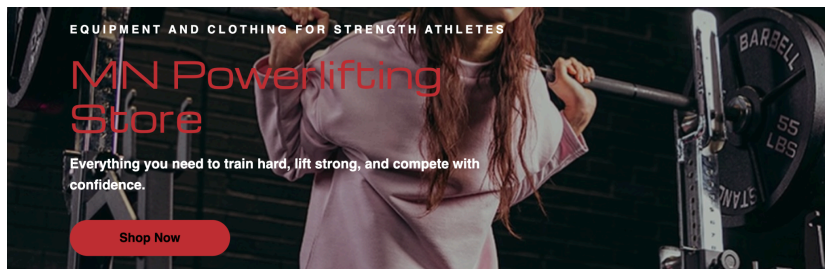
The site makes options visible instead of asking users to remember them. The All Products dropdown shows Equipment, Clothing, and the subcategories directly. Product cards also show category, type, variation, focus, and price, so users do not need to remember product information from another page. This makes browsing easier and reduces memory load.

Shop by Training Need

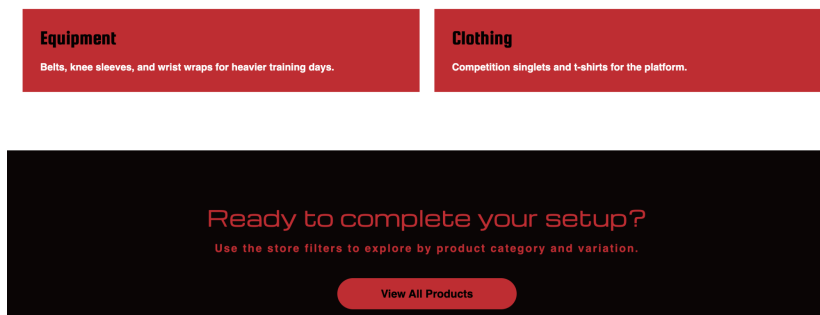


7. Flexibility and Efficiency of Use

Users can browse the store in different ways depending on their needs. A beginner can click Shop Now and view all products, while a user who knows what they want can go directly to Equipment, Belts, Knee Sleeves, Wrist Wraps, Competition Singlets, or Competition T-Shirts. This supports both slow exploration and faster shopping.



Shop by Training Need



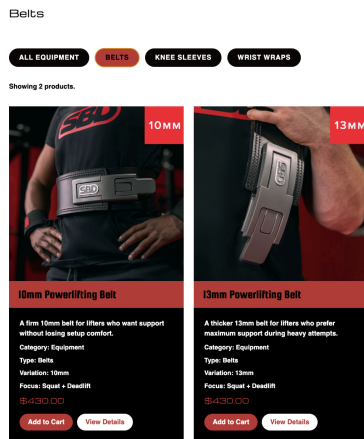
Ready to complete your setup?

Use the store filters to explore by product category and variation.

View All Products

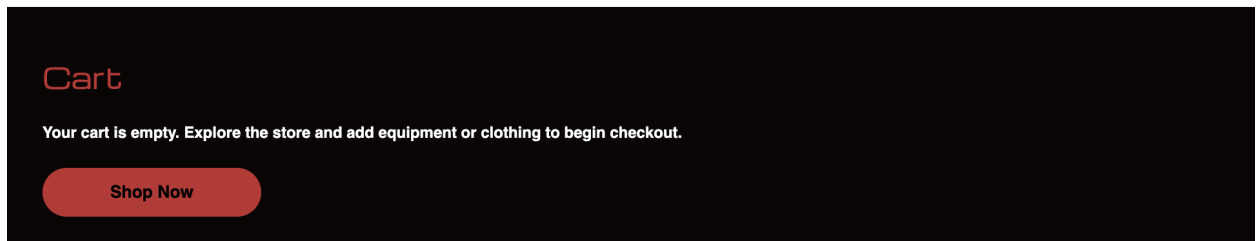
8. Aesthetic and Minimalist Design

The site keeps the interface focused on the shopping task. I removed the search box and used a simple category-based navigation system instead. The product cards show only the most important information: image, name, description, category, type, variation, focus, price, and action buttons. This avoids unnecessary clutter and keeps the design clean.



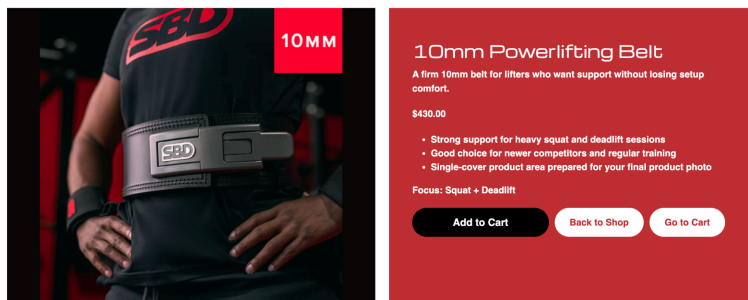
9. Help Users Recognize, Diagnose, and Recover from Errors

When the cart is empty, the site clearly tells the user that the cart is empty and provides a Shop Now button to recover from that state.



10. Help and Documentation

The site provides enough guidance for users to complete the shopping task without needing external help. Product cards and product detail pages explain what each product is for, including its category, variation, focus, and price. The checkout labels clearly tell users what information to enter.



Generative AI Acknowledgement

I used generative AI as a support tool during this assignment, but I used it differently for the prototype and the report.

For the high-fidelity prototype, AI was most helpful with the React development and debugging process. I used ChatGPT to help plan and revise the interactive structure of the e-commerce site, including the product list, category-based faceted search, product detail pages, shopping cart, checkout flow, purchase confirmation, purchase history, and post-checkout survey. AI also helped me debug issues related to image display, button styling, form validation, checkout steps, React state management, and navigation between pages. However, I tested and adjusted the code myself to make sure the final prototype matched my intended design and worked correctly.

For the visual design, I did not use AI to create the design directly. I chose the powerlifting store topic, product categories, black/red/white color palette, typography, layout style, and overall brand direction myself. Since the site is connected to my MN Powerlifting theme, I made the design decisions based on my own knowledge of powerlifting and my previous portfolio/service site style. AI helped me organize some design explanations, but the actual visual style and final design choices were made by me.

For the report, AI helped me organize my ideas and improve the clarity of my writing. I used ChatGPT to help explain the e-commerce goal, inspiration sites, interactive processes, verbal communication choices, visual design choices, heuristic evaluation, and the role of React in the prototype. I also used AI to check whether my website and report matched the assignment requirements. The final wording, testing, design decisions, and submitted work were reviewed and edited by me.